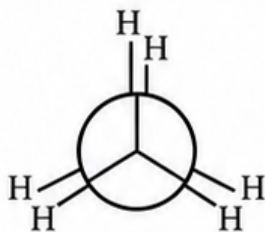


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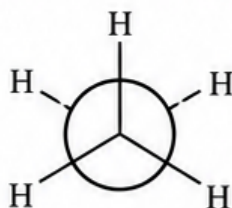
40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q1. Number of eclipsed conformations obtained in one complete rotation of ethane = ____.



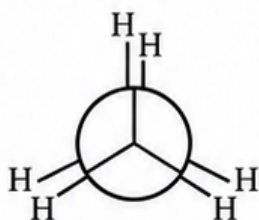
Eclipsed conformation of ethane

Q2. Number of staggered conformations obtained in one complete rotation of ethane = ____.

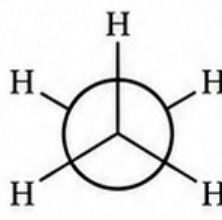


Staggered conformation of ethane

Q3. Most stable conformer of ethane is ____.

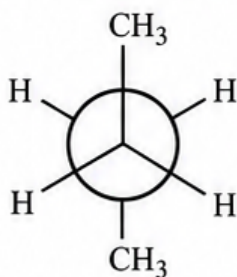


Eclipsed

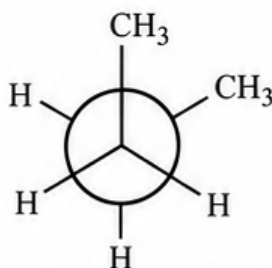


Staggered

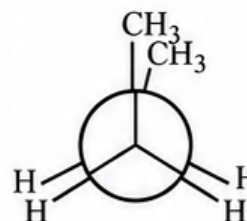
Q4. Most stable conformer of *n*-butane is ____.



Anti

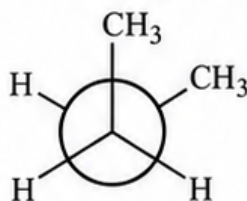


Gauche



Fully eclipsed

Q5. Number of gauche conformers of *n*-butane in one complete rotation = ____.

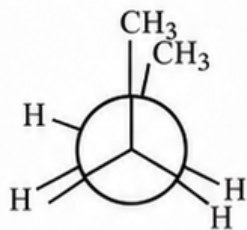


Gauche conformation of *n*-butane

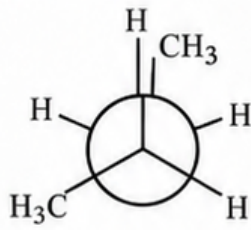
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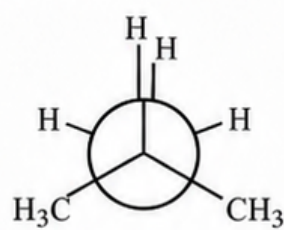
Q6. Number of eclipsed conformers of *n*-butane in one complete rotation = _____.



Eclipsed

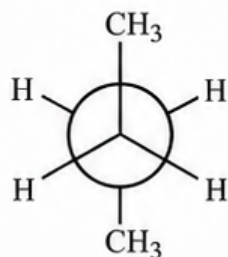


Eclipsed

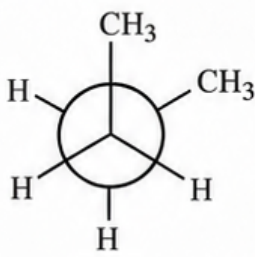


Eclipsed

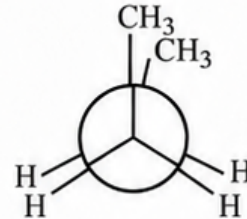
Q7. Highest-energy conformer of *n*-butane is _____.



Anti

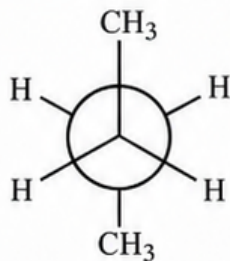


Gauche

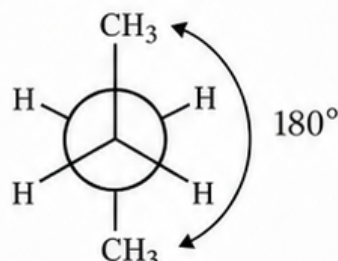


Fully eclipsed

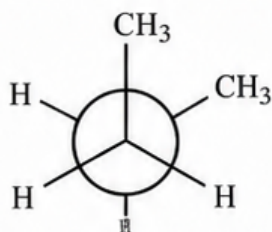
Q8. Dihedral angle between the two methyl groups in *anti n*-butane = _____.



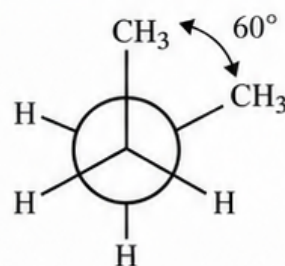
Anti



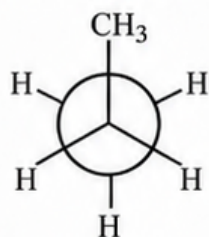
Q9. Dihedral angle between the two methyl groups in *gauche n*-butane = _____.



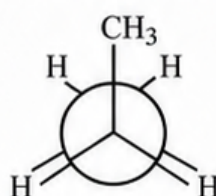
Gauche



Q10. Number of distinct conformers of *propane* (one staggered type and one eclipsed type) = _____.



Staggered (all H)



Eclipsed (all H)

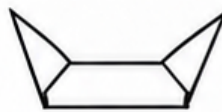
CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q11. Most stable conformation of cyclohexane is ____.



Chair



Boat

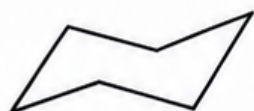


Twist-boat

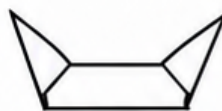


Half-chair

Q12. Least stable common conformer of cyclohexane is ____.



Chair



Boat

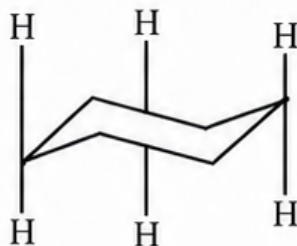


Twist-boat

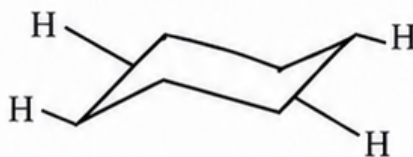


Half-chair

Q13. Number of axial positions in one chair conformer of cyclohexane = ____.



Q14. Number of equatorial positions in one chair conformer of cyclohexane = ____.

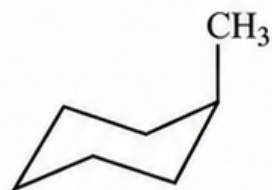


Q15. On ring flip, axial bonds become ____ bonds.

CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q16. Preferred position of the methyl group in methylcyclohexane = ____.

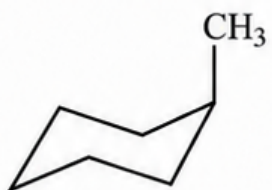


Axial methylcyclohexane



Equatorial methylcyclohexane

Q17. Less stable conformer of methylcyclohexane has CH_3 in ____ position.

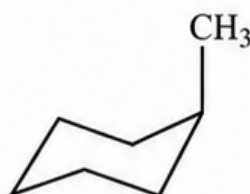


Axial methylcyclohexane
(less stable)



Equatorial methylcyclohexane
(more stable)

Q18. Number of 1,3-diaxial interactions in axial methylcyclohexane = ____.



Axial methylcyclohexane

Q19. Preferred position of the *tert*-butyl group in *tert*-butylcyclohexane = ____.



Axial *tert*-butylcyclohexane



Equatorial *tert*-butylcyclohexane

Q20. More stable conformer of fluorocyclohexane has F in ____ position.



Axial fluorocyclohexane

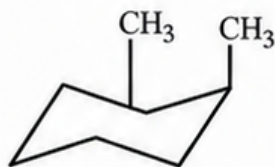


Equatorial fluorocyclohexane

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40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q21. *cis*-1,2-dimethylcyclohexane in a chair is ____ (diaxial / diequatorial / one axial and one equatorial).

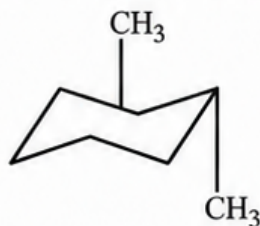


cis-1,2-dimethylcyclohexane
(diaxial)

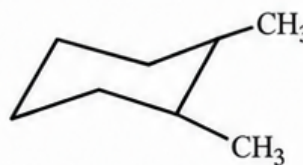


cis-1,2-dimethylcyclohexane
(diequatorial)

Q22. The more stable conformer of *trans*-1,2-dimethylcyclohexane is ____.

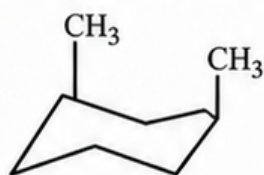


trans-1,2-dimethylcyclohexane
(diaxial)

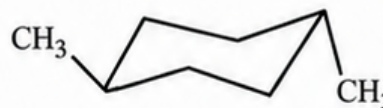


trans-1,2-dimethylcyclohexane
(diequatorial)

Q23. The more stable conformer of *cis*-1,3-dimethylcyclohexane is ____.

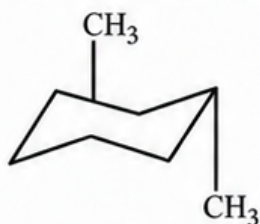


cis-1,3-dimethylcyclohexane
(diaxial)

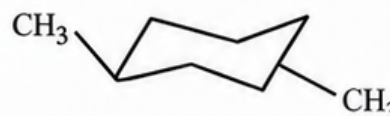


cis-1,3-dimethylcyclohexane
(diequatorial)

Q24. *trans*-1,3-dimethylcyclohexane in a chair is ____ (diaxial / diequatorial / one axial and one equatorial).

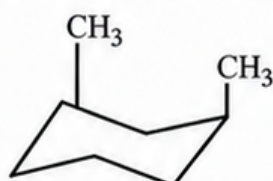


trans-1,3-dimethylcyclohexane
(diaxial)

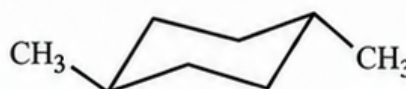


trans-1,3-dimethylcyclohexane
(diequatorial)

Q25. *cis*-1,4-dimethylcyclohexane in a chair is ____ (diaxial / diequatorial / one axial and one equatorial).



cis-1,4-dimethylcyclohexane
(diaxial)

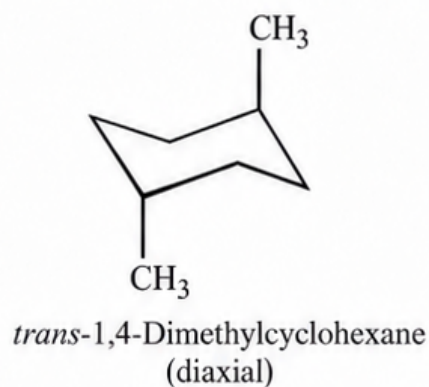
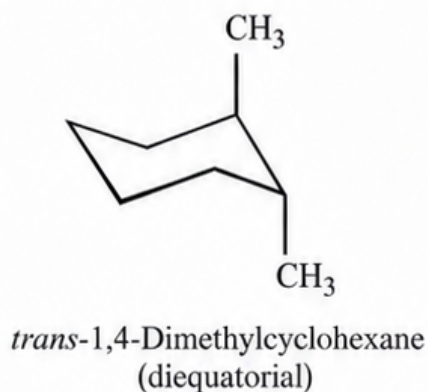


cis-1,4-dimethylcyclohexane
(diequatorial)

CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q26. The more stable conformer of *trans*-1,4-dimethylcyclohexane is _____.

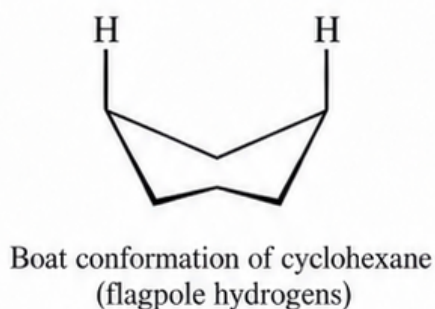


Q27. Ring flip _____ configuration of a substituted cyclohexane (changes / does not change).

Q28. Axial C-X bonds in cyclohexane are oriented _____ to the average ring axis.

Q29. Number of equivalent chair conformers of cyclohexane related by ring flip = _____.

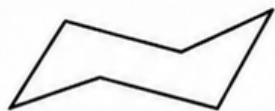
Q30. The interaction that destabilizes boat cyclohexane is called _____ interactions.



CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

Q31. Arrange the following conformers in decreasing stability: chair, twist-boat, boat, half-chair.



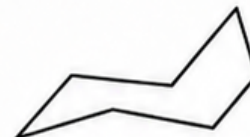
Chair



Twist-boat



Boat



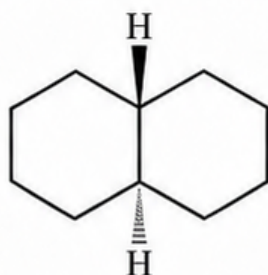
Half-chair

Q32. Anti *n*-butane is more stable than gauche *n*-butane by about ____ kcal mol⁻¹.

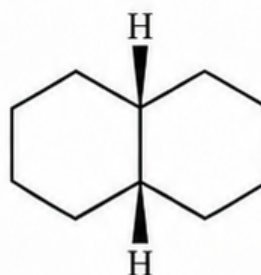
Q33. Barrier to rotation of ethane is about ____ kcal mol⁻¹.

Q34. Larger substituents on cyclohexane prefer the ____ position.

Q35. Which fused bicyclic system undergoes ring flip more readily: *cis*-decalin or *trans*-decalin? ____.



cis-decalin

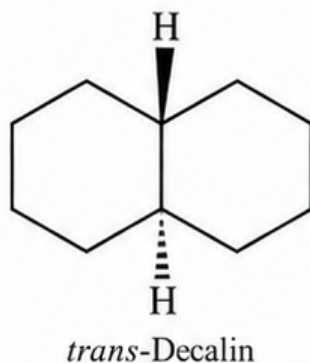


trans-decalin

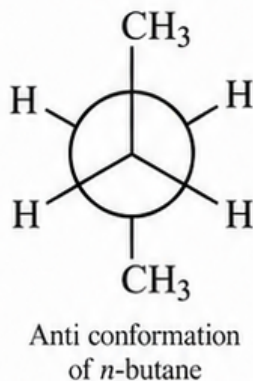
CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

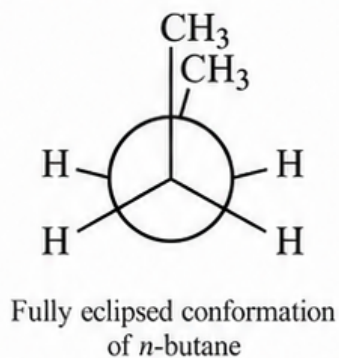
Q36. *trans*-Decalin is conformationally _____.



Q37. Number of *anti* conformers of *n*-butane in one complete rotation = _____.



Q38. Number of fully eclipsed CH₃-CH₃ conformers of *n*-butane in one complete rotation = _____.



Q39. Axial substituents in cyclohexane experience _____ interactions.

Q40. Conformational isomerism mainly arises due to rotation about _____ bonds.

CONFORMATIONAL ISOMERISM ASSIGNMENT

40 QUESTIONS | CONFORMATIONAL ISOMERISM ONLY

ANSWER KEY

Q1	3	Q21	one axial and one equatorial
Q2	3	Q22	diequatorial
Q3	staggered	Q23	diequatorial
Q4	anti	Q24	one axial and one equatorial
Q5	2	Q25	one axial and one equatorial
Q6	3	Q26	diequatorial
Q7	fully eclipsed	Q27	does not change configuration
Q8	180°	Q28	parallel to the ring axis
Q9	60°	Q29	2
Q10	2	Q30	flagpole
Q11	chair	Q31	chair > twist-boat > boat > half-chair
Q12	half-chair	Q32	0.9
Q13	6	Q33	3.0
Q14	6	Q34	equatorial
Q15	equatorial	Q35	cis-decalin
Q16	equatorial	Q36	locked
Q17	axial	Q37	1
Q18	2	Q38	1
Q19	equatorial	Q39	1,3-diaxial
Q20	equatorial	Q40	single (sigma)

Conformational isomerism only; configurational isomerism is excluded unless explicitly stated.